

DESIGN AND ANALYSIS OF ALGORITHM

MOODLE PROBLEM SOLVING

WEEK 1:

Q)1

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
{
    int i= 1;

    int x =1;

    while(x <= n)
    {
        i++;
        x = i;
    }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

Ace editor not ready. Perhaps reload page?

Falling back to raw text area.

```
int i= 1;
count++;
int x =1;
count++;
while(x <= n)
{
    count++;
    i++;
    count++;
    x = i;
    count++;
}
count++;
printf("%d",count);
}
```

	Input	Expected	Got	
✓	9	12	12	✓
✓	4	9	9	✓

Passed all tests: ✓

Q)2

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
        printf("%d");
    }
    else
    {
        for(int i=1; i<n; i++)
        {
            for(int j=1; j<n; j++)
            {
                printf("%d");
                printf("%d");
                break;
            }
        }
    }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer: (penalty regime: 0 %)

```
1 #include <stdio.h>
2 int main(){
3     int n;
4     scanf("%d",&n);
5     int count=0;
6
7     if(n==1){
8         count++;
9     }
10    else{
11        count+=2;
12
13        for(int i=1; i<n; i++){
14            count++;
15
16            for(int j=1; j<n; j++){
17                break;
18            }
19        }
20    }
21    printf("%d", count);
22    return 0;
23 }
24
25
26
27
28
29
```

	Input	Expected	Got	
✓	3	12	12	✓
✓	1000	1002	1002	✓
✓	143	717	717	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00

Q)3

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {  
    for (i = 1; i <= num; ++i)  
    {  
        if (num % i == 0)  
        {  
            printf("%d ", i);  
        }  
    }  
}
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:
A positive Integer n

Output:
Print the value of the counter variable

Answer:

```
1 #include <stdio.h>  
2  
3 void Factor(int num){  
4     int count=0;  
5  
6     for(int i=1; i<=num; ++i){  
7         count++;  
8         if(num%i==0){  
9             count++;  
10        }  
11    }  
12    count++;  
13    printf("%d", count);  
14 }  
15  
16 int main()  
17 {  
18     int n;  
19     scanf("%d", &n);  
20     Factor(n);  
21     return 0;  
22 }  
23  
24  
25
```

Q)4

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c=0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1 #include <stdio.h>
2 int main()
3 {
4     int n;
5     scanf("%d", &n);
6     int c=0;
7     c++;
8
9     for(int i=n/2; i<n; i++){
10         c++;
11         for(int j=1; j<n; j*=2){
12             c++;
13             for(int k=1; k<n; k*=2){
14                 c++;
15                 c++;
16             }
17             c++;
18         }
19     }
20     c++;
21
22 }
23 c++;
24 printf("%d", c);
25 return 0;
26 }
```

Q)5

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
    int rev = 0, remainder;
    while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n /= 10;
    }
    printf("%d", rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1 #include <stdio.h>
2 int main(){
3     int n;
4     scanf("%d", &n);
5     int c=0;
6     c++;
7     int rev=0, remainder;
8     c++;
9
10    while(n!=0){
11        c++;
12        remainder=n%10;
13        c++;
14        rev=rev*10+remainder;
15        c++;
16        n/=10;
17        c++;
18    }
19    c++;
20    printf("%d", c);
21    return 0;
22 }
```